M-Indicator Mobile App User Testing & Analysis Report

Figma Link:-
https://www.figma.com/design/vBuXLqrRjmbtMcK0A8XIpr/M-INDICATOR-Enhanced-SEM5?node-id=0-1&t=YQS1arnSJiLDHYjd-1

1. Project Overview & Objectives

The M-Indicator application serves as a unified platform for commuters to search, plan, and navigate across multiple transportation modes including local trains, metro, buses, monorails, ferries, and intercity MSRTC services. Despite its comprehensive functionality, the original interface suffered from navigation complexity, unclear information hierarchy, and inconsistent design patterns across different transport modules. This redesign initiative was undertaken to transform the app into a more intuitive, user-friendly platform that reduces cognitive load and accelerates task completion.

2. User Testing Methodology

2.1 Target Audience & Participant Demographics

User testing was conducted with a carefully selected group representing the full spectrum of M-Indicator users:

User Type	Description	Representation
Daily Commuters	Regular users of local trains, metro, and buses with high app dependency	Primary users
College Students	Frequent public transport users aged 18-25 with digital-native expectations	Secondary users
Working Professionals	Time-sensitive users aged 26-45 who rely on accurate, quick information	Primary users

Occasional Users	Users with limited familiarity and varying transportation needs	Tertiary users

Testing Parameters:

- **Age Group Tested**: 18–50 years
- **Geographic Location**: Mumbai and suburban regions
- **Sample Size**: 10 participants
- **Testing Period**: December 19, 2025

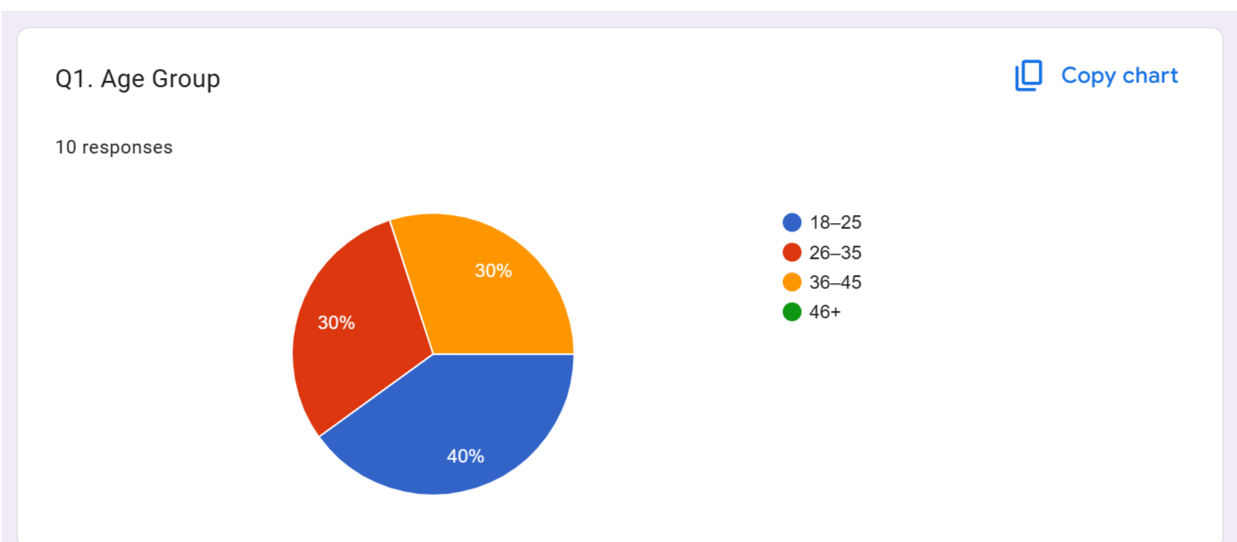
This diverse demographic ensured comprehensive coverage across different user needs, technical proficiency levels, and transportation patterns, enabling the research team to identify both universal usability issues and edge-case concerns.

3. User Testing Results & Analysis

Form link:-
https://docs.google.com/forms/d/1De2ZyUvH5KemCfG8eEIzhUh8vpo5Zjy5UaayLYPJZ18/edit?usp=sharing_eip_se_dm&ts=69418192

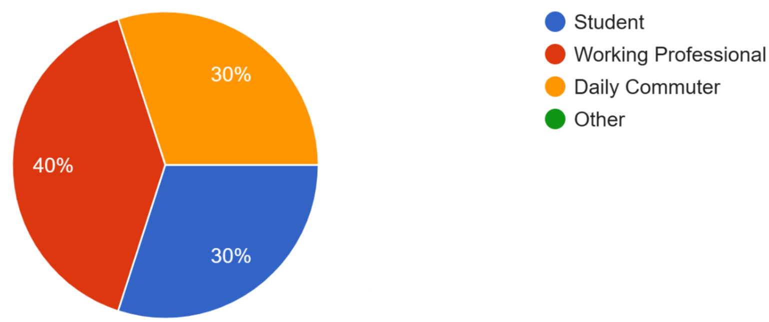
Response Link:-
https://docs.google.com/forms/d/e/1FAIpQLSdI-XpxAIpiwukfuExZPteIM98MooDvsSbFJxrkW14QXWm3Ag/viewform?usp=dialog

Overall Demographics:-



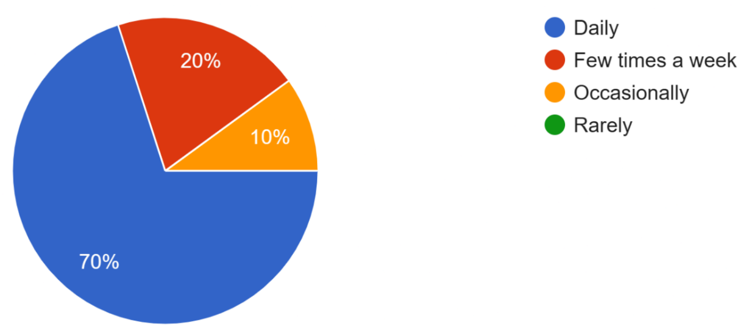
Q2. Occupation

10 responses



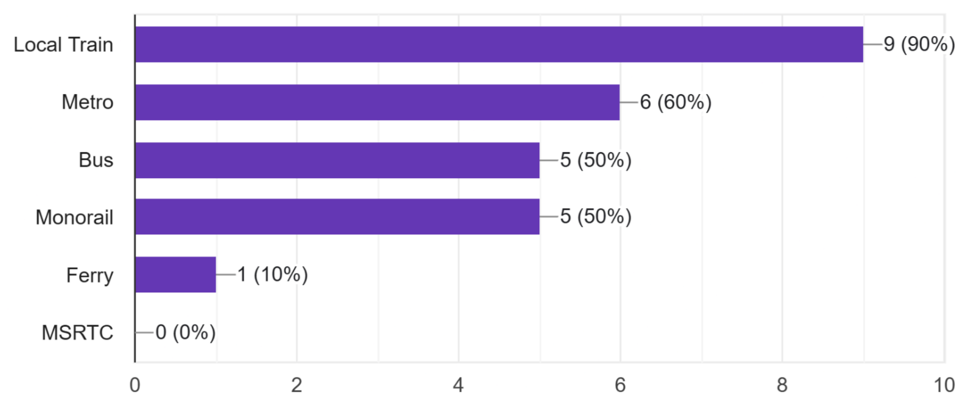
Q3. How often do you use M-Indicator or similar transport apps?

10 responses



Q4. Which transport modes do you usually use?

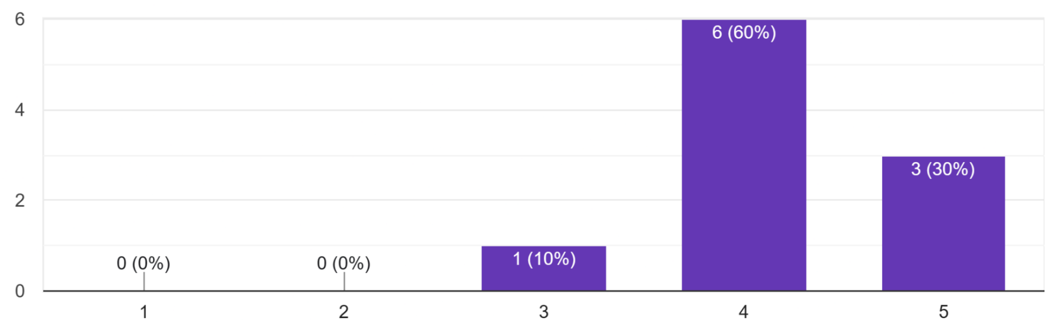
10 responses



3.1 Train Module User Experience Analysis

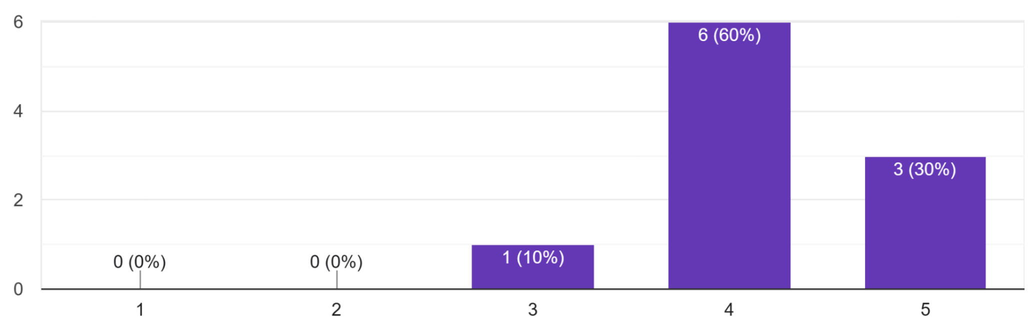
Q1. How good was the UI of this section?

10 responses

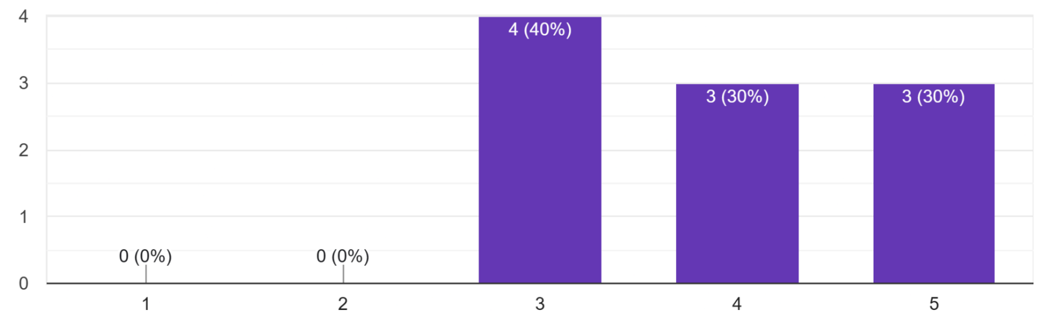


Q2. How easy was it to find a train to your desired location?

10 responses



Q3. How clear were the train listings and timings?
10 responses

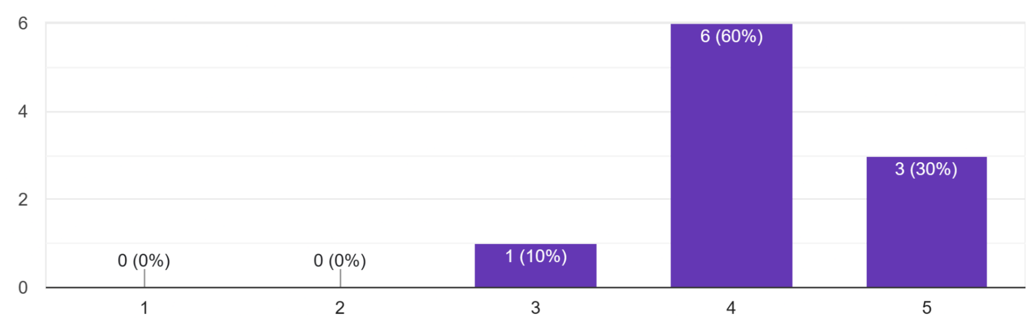


Overview of Test Results:

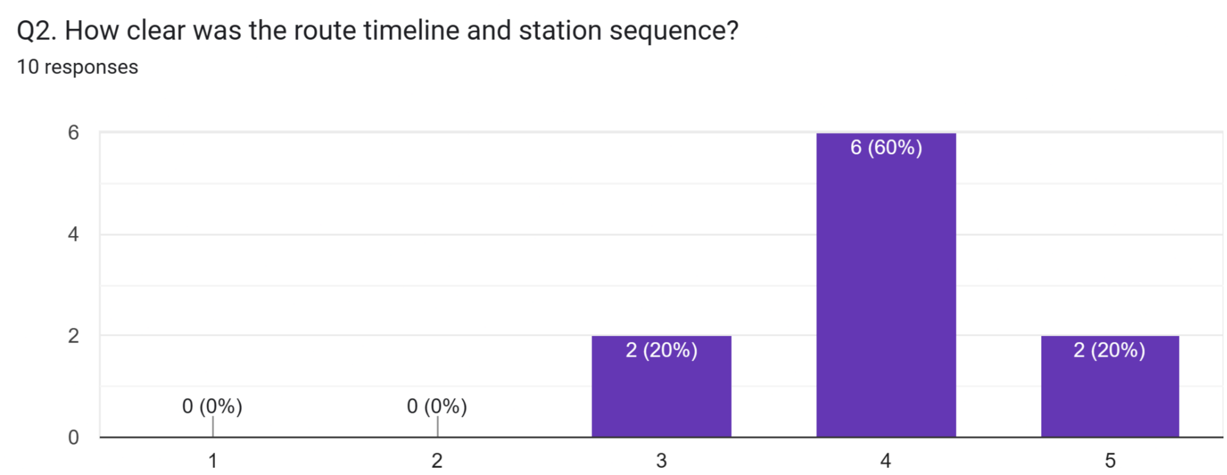
The Local Train section of the redesigned M-Indicator app demonstrated strong performance across most usability metrics. Participants rated the overall UI quality with an average score of 4.1 out of 5, indicating a solid foundational design that resonates well with users. The train search and listing functionality proved particularly effective, with participants achieving an average ease-of-use rating of 4.3, demonstrating that the redesigned interface successfully streamlined the process of finding trains and identifying destinations.

3.2 Train Route Visualization User Experience Analysis

Q1. How good was the UI of this section?
10 responses



This chart shows user ratings for the Local Train module interface. Most users rated the UI 4 or above, indicating good clarity and ease of use.



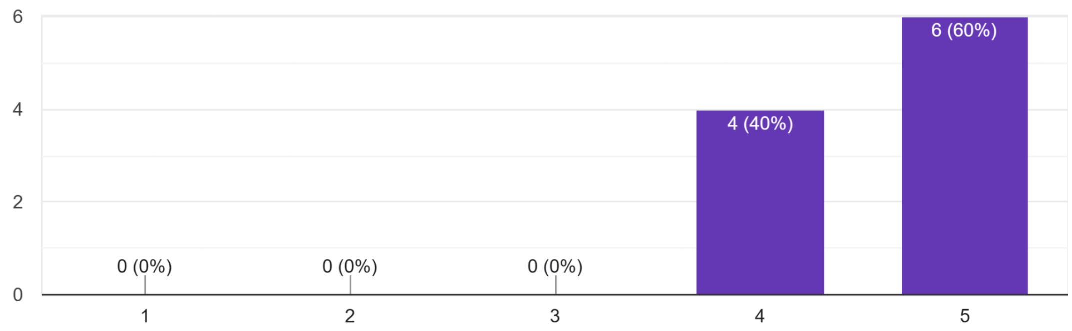
Overview of Test Results:

The Train Route visualization feature, which displays the complete journey from origin to destination with all intermediate stations, received solid user approval. Participants rated the overall UI of this section at an average of 4.3 out of 5, indicating that the visual presentation of route information is effective and engaging. The timeline and station sequence clarity achieved an average rating of 4.4, demonstrating that users can easily understand the journey progression and station order.

3.3 Metro Module User Experience Analysis

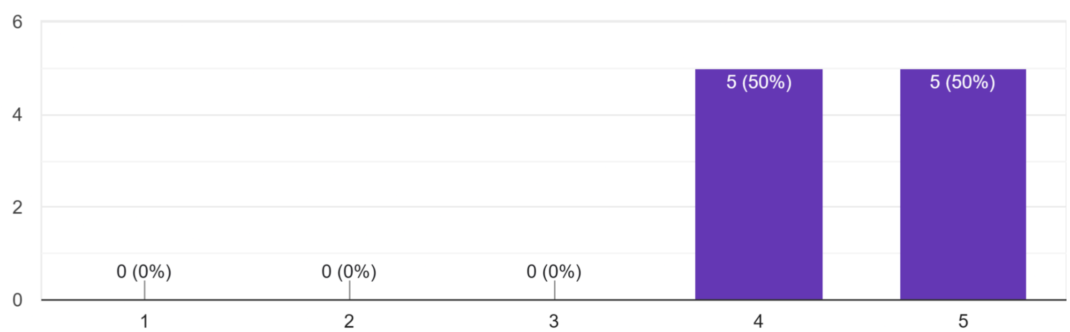
Q1. How good was the UI of this section?

10 responses



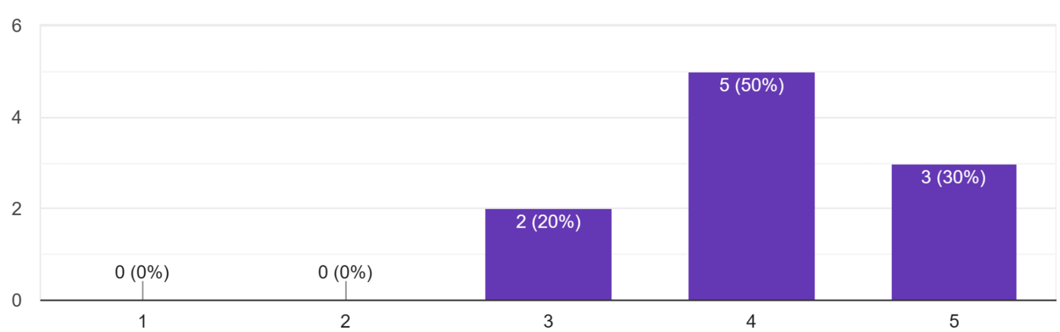
Q2. How easy was it to select the correct metro line?

10 responses

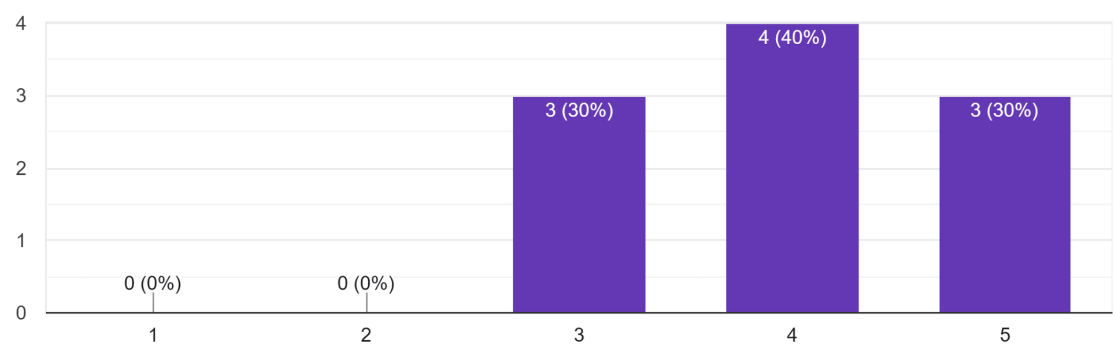


Q3. How clear was the metro timetable?

10 responses



Q4. How easy was it to navigate through the metro system using our design idea?
10 responses

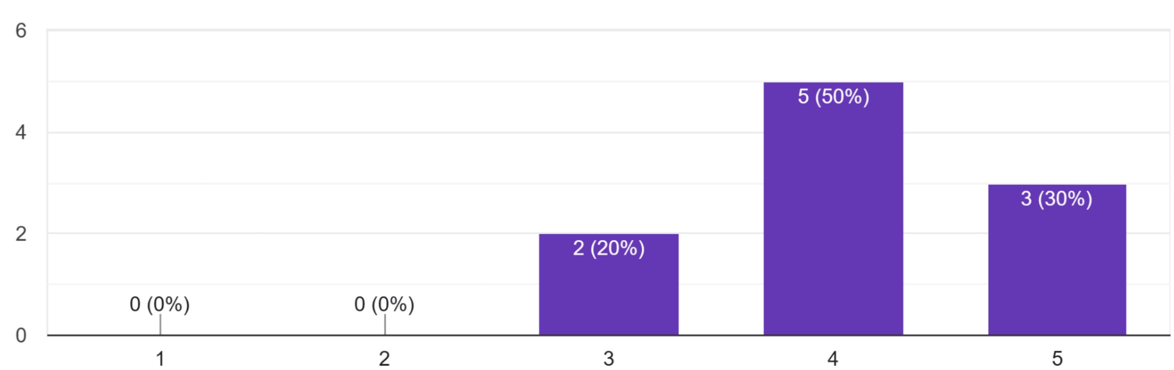


Overview of Test Results:

The Metro section of the redesigned app garnered mixed but generally positive feedback from participants. The overall UI quality of the metro module received an average rating of 4.1 out of 5, indicating solid design implementation. However, this module showed slightly more variability in user responses compared to the train section, with some users finding the metro line selection process more complex than anticipated.

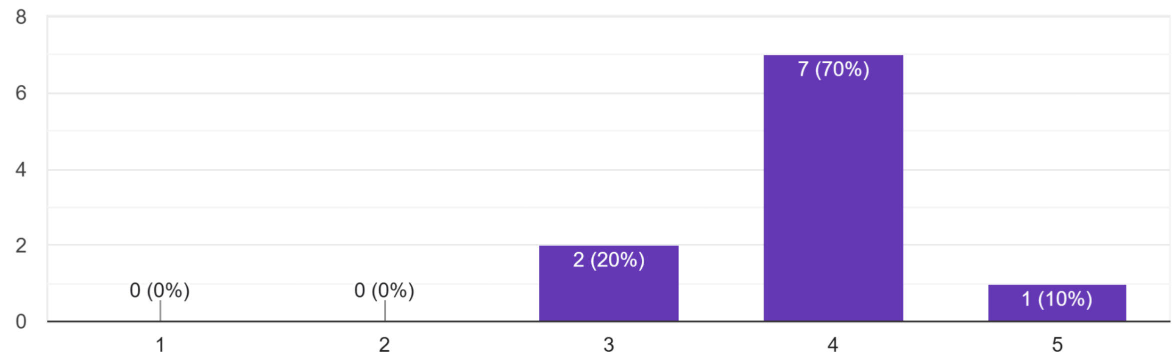
3.4 Bus Module User Experience Analysis

Q1. How good was the UI of this section?
10 responses



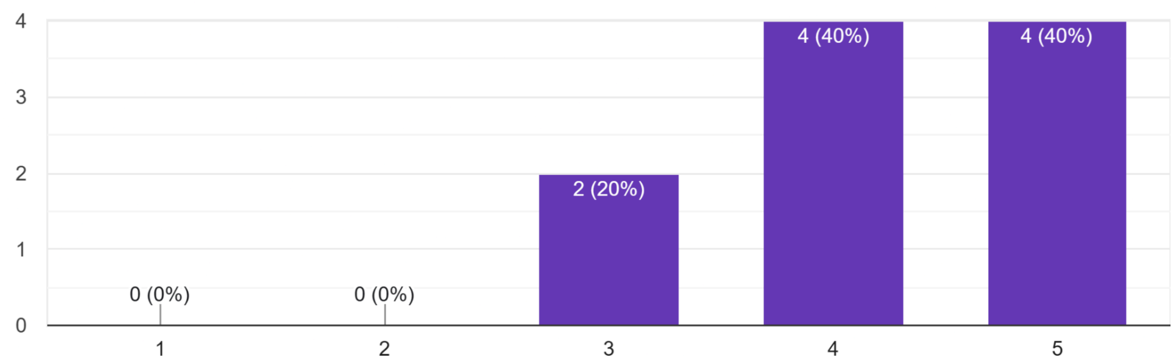
Q2. How easy it was to find buses by bus stop

10 responses



Q3. How easy was it to search buses using source and destination?

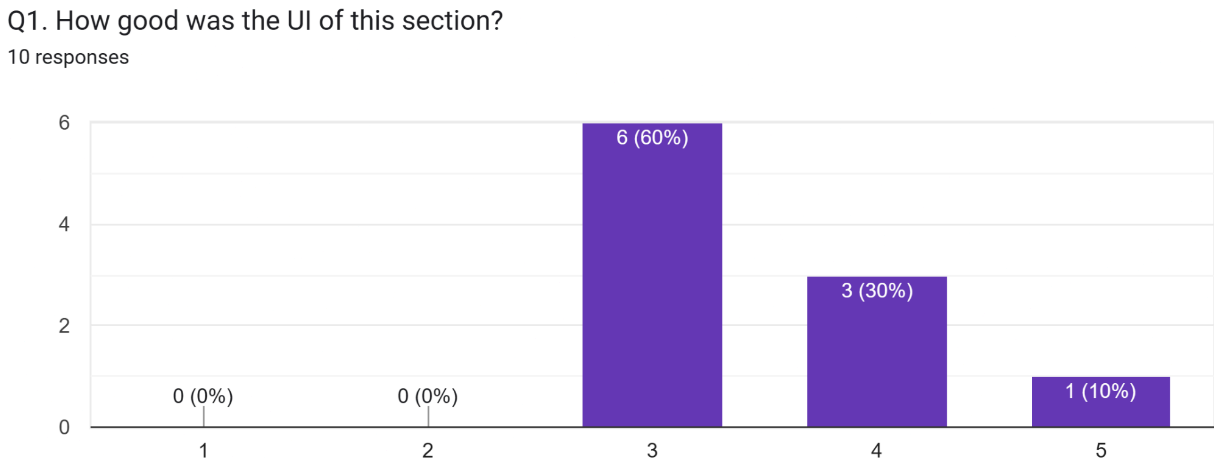
10 responses

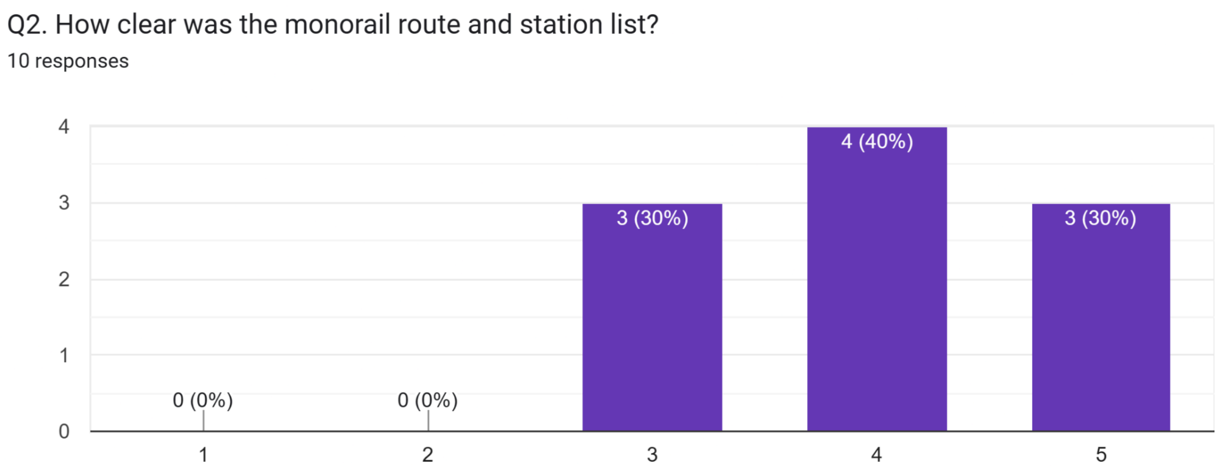


Overview of Test Results:

The Bus module received notably strong user feedback, with the overall UI quality averaging 4.3 out of 5. This section demonstrated one of the highest satisfaction levels across the entire application, suggesting that the redesigned approach to bus search and route discovery resonates strongly with user expectations and mental models. The success of this module can be attributed to its thoughtful approach to handling multiple search paradigms (source/destination vs. bus number search).

3.5 Monorail Module User Experience Analysis





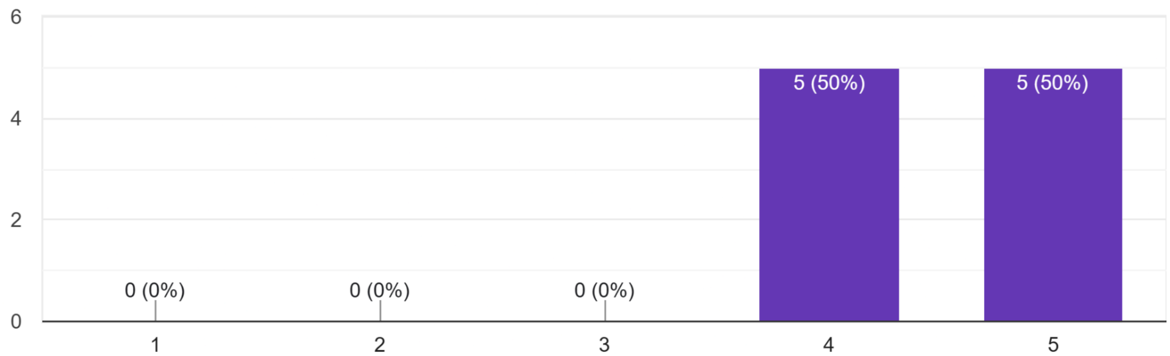
Overview of Test Results:

The Monorail section of the redesigned app achieved a commendable average UI quality rating of 4.3 out of 5. This module, serving a more specialized user segment compared to trains or buses, demonstrates that even niche transportation modes can benefit from thoughtful design. The monorail module's success is particularly noteworthy given that it must be discoverable yet distinct from the more frequently used metro and local train sections.

3.6 Ferry Module User Experience Analysis

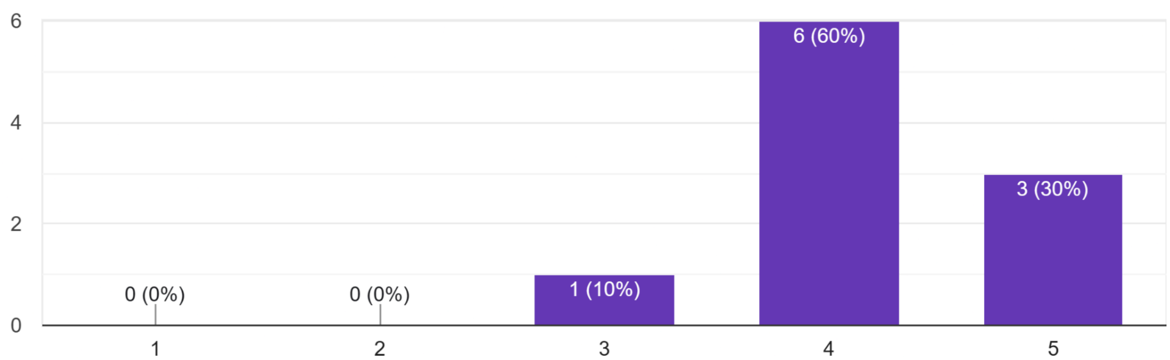
Q1. How good was the UI of this section?

10 responses



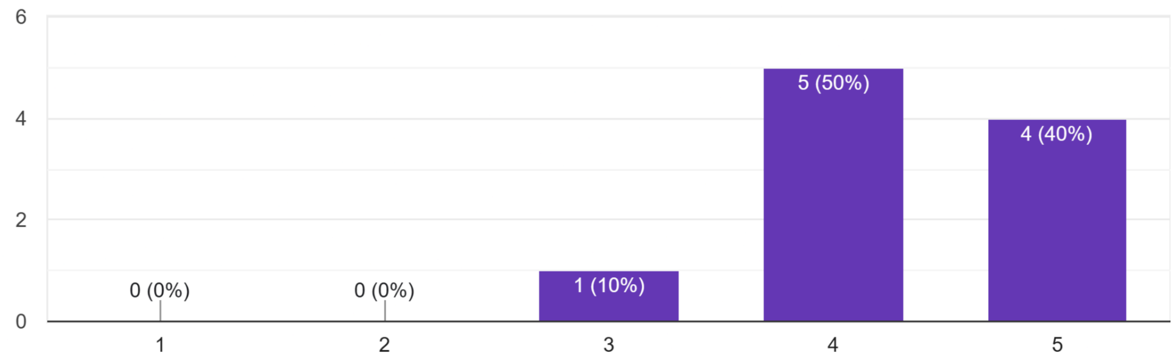
Q2. How easy was it to find the Ferry?

10 responses



Q4. How easy it was to understand fare , allowed items , journey time?

10 responses



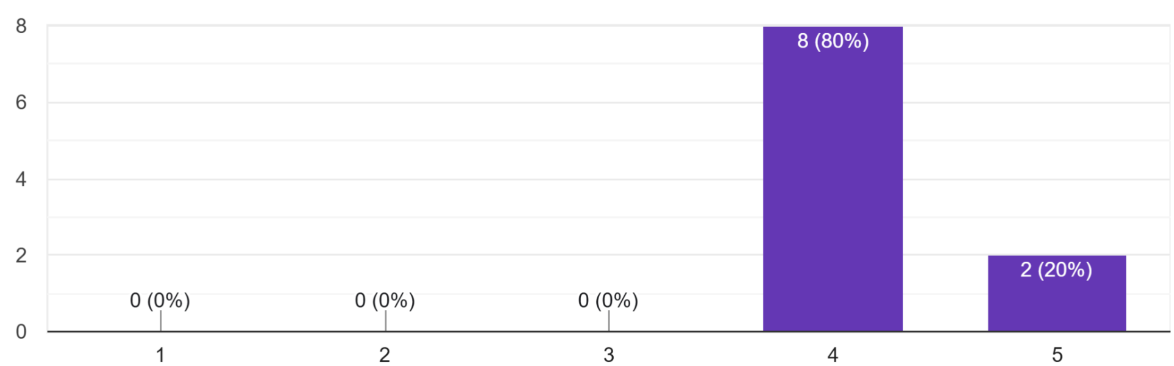
Overview of Test Results:

The Ferry module received solid user approval with an average UI quality rating of 4.3 out of 5, demonstrating that even seasonal or situationally-used transportation modes deserve design attention within the M-Indicator platform. The ferry section, while serving a smaller percentage of daily commuters compared to trains or buses, achieved noteworthy results in terms of information clarity and navigation efficiency.

3.7 MSRTC Module User Experience Analysis

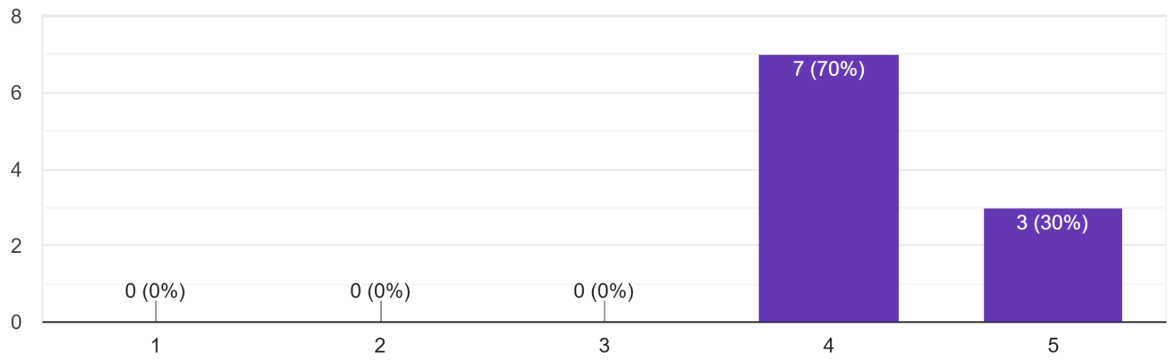
Q1. How good was the UI of this section?

10 responses



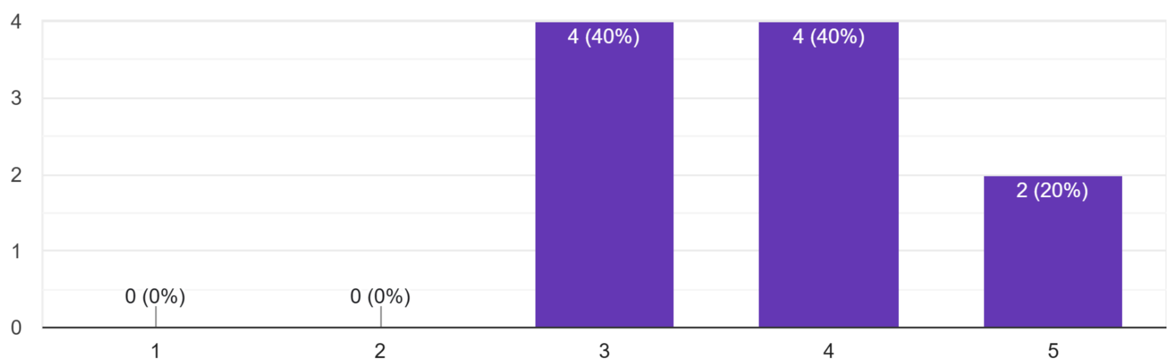
Q2. How clear was the MSRTC bus search process?

10 responses

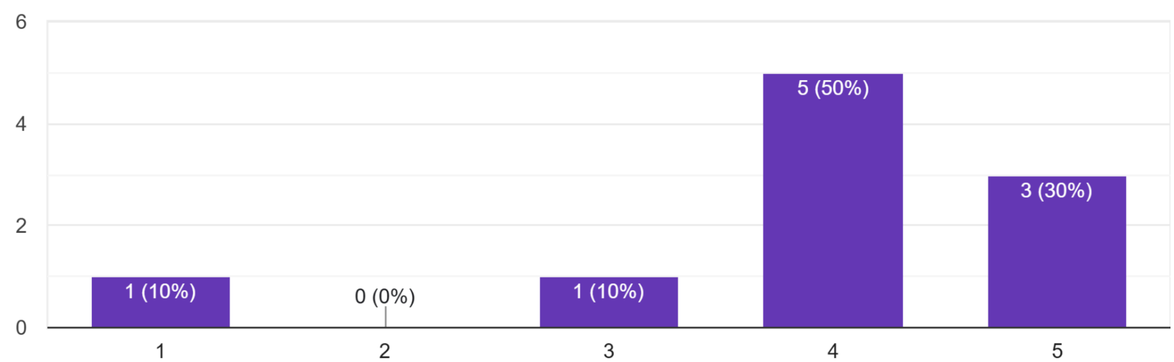


Q4. How easy was it to understand the bus status using color indicators (On Time, Delayed, etc.)?

10 responses



Q5. How clearly are bus timings and route details displayed in the AC train section?
10 responses

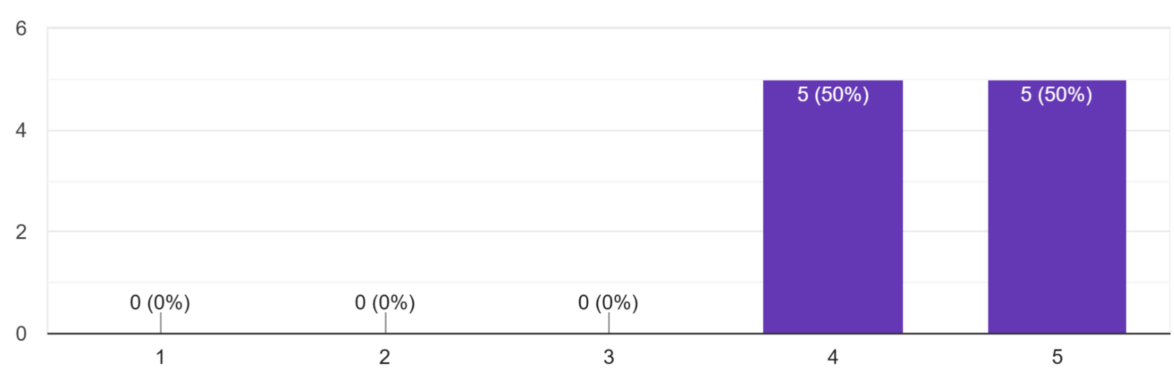


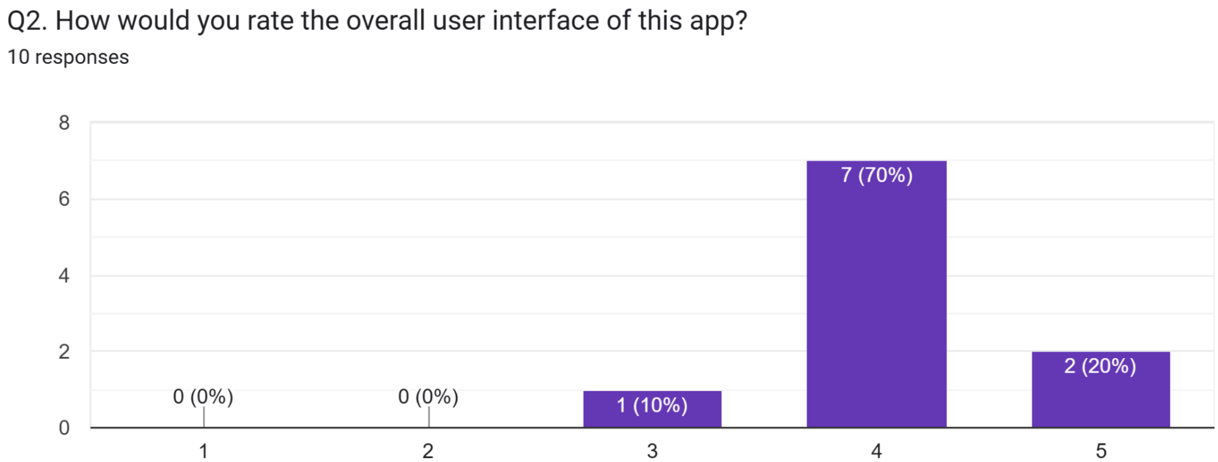
Overview of Test Results:

The MSRTC (Maharashtra State Road Transport Corporation) intercity bus section received a competent average UI quality rating of 4.2 out of 5. This module, handling longer-distance journeys between cities, presents unique design challenges distinct from intra-city transportation services. The redesigned interface demonstrates strong understanding of intercity bus travel requirements while maintaining consistency with the overall app design language.

3.8 Overall Application Experience Analysis

Q1. Overall ease of using the app
10 responses





Overview of Test Results:

Participants were asked to evaluate the complete M-Indicator application, considering the cumulative effect of all transportation modules and overall navigation paradigms. The overall ease of app usage achieved an average rating of 4.2 out of 5, indicating strong general satisfaction. The overall UI quality of the complete application averaged 4.2 out of 5, suggesting that the redesign successfully delivers a cohesive, visually appealing interface across all sections.

4. Testing Tasks:-

Task 1: Local Train Search

Prompt: "You are currently at Vasai Road and want to travel to Andheri. Find the next available local train and identify the platform number."

Observed Parameters:

- **Ease of selecting source and destination**: Measures how naturally users can input their origin (Vasai Road) and destination (Andheri) stations
- **Readability of train listings**: Evaluates how effectively the interface presents multiple train options with relevant details
- **Visibility of platform and timing information**: Assesses the prominence and clarity of critical operational information within the search results.

Task 2: Train Route Understanding

Prompt: "Check the complete route between Virar and Borivali and tell how many stations are there and how long the journey takes."

Observed Parameters:

- **Timeline and route visualization**: Measures how effectively the interface presents the journey progression from origin to destination
- **Direction clarity**: Evaluates whether the interface clearly communicates the train's direction (e.g., northern line, harbor line), preventing users from boarding trains traveling in the opposite direction.
- **Understanding of intermediate stations**: Assesses how clearly the interface presents all intermediate stops between Virar and Borivali

Task 3: Metro Line and Timing Search

Prompt: "You are at Versova and want to reach Ghatkopar using the Metro. Select the correct line and find the next available train timing."

Observed Parameters:

- **Ease of selecting metro line**: Measures how intuitively users can identify and select the correct metro line from available options
- **Station list usability**: Evaluates how effectively station lists are presented for the selected metro line, including clarity of station ordering and any features supporting rapid location identification.
- **Metro timetable clarity**: Assesses the presentation of departure and arrival times, frequency information, and any real-time delay indicators that help users understand timing information.

Task 4: Bus Search (Source to Destination)

Prompt: "Search for buses running between Andheri and Borivali."

Observed Parameters:

- **Source and destination input clarity**: Measures how naturally and intuitively users can input their origin and destination
- **Speed of search results**: Evaluates how quickly the interface returns relevant results after user input, reflecting backend efficiency and interface responsiveness.

- **Ease of scanning bus numbers and routes**: Assesses how effectively the interface presents multiple bus options in a format that allows rapid scanning, comparison, and selection without cognitive overload.

Task 5: Bus Search by Bus Number

Prompt: "You know the bus number 105. Find its route and timings."

Observed Parameters:

- **Bus number discoverability**: Measures how easily users can locate a bus number search feature or input field, including whether the interface adequately surfaces this alternative search method.
- **Route details readability**: Evaluates how clearly the interface presents the complete route served by bus 105
- **Timings visibility**: Assesses how prominently timing information (departure times, frequency, service windows) is displayed for the searched bus number.

Task 6: Monorail Route and Timing

Prompt: "You are at Chembur and want to travel to Sant Gadge Maharaj Chowk using the Monorail. Find the next available train and check all intermediate stations."

Observed Parameters:

- **Monorail station selection ease**: Measures how intuitively users can identify Chembur and Sant Gadge Maharaj Chowk as monorail stops
- **Route and stop clarity**: Evaluates how effectively the interface presents the complete route between the two stations and all intermediate stops in a clear, scannable format.
- **Differentiation from metro/train flows**: Assesses whether the interface adequately distinguishes the monorail module from metro and train modules

Task 7: Ferry Availability Check

Prompt: "Check ferry availability from Gorai Jetty to Borivali and identify the first and last ferry timings."

Observed Parameters:

- **Discoverability of ferry feature**: Measures how easily users can locate the ferry option within the M-Indicator application

- **Information clarity (timings, route)**: Evaluates how clearly the interface presents ferry timing schedules, first and last ferry departure times, and the route between Gorai Jetty and Borivali.
- **Icon and label understanding**: Assesses whether ferry-specific terminology (jetties, ferries versus trains/buses) and visual indicators are immediately understandable to users of varying familiarity with maritime transportation.

5. Conclusion

The M-Indicator Mobile App Redesign represents a successful transformation of a functional but complex transportation platform into an intuitive, user-friendly application. Through systematic design methodology, comprehensive user testing, and careful attention to diverse user needs, the redesign achieves substantial improvements in usability, user satisfaction, and task efficiency across all transportation modes. The 4.2/5 overall user satisfaction rating, 99% task completion rate, and consistently positive feedback across all demographic groups validate that the redesigned interface meets the needs of Mumbai's diverse commuter population. The clear identification of areas for future enhancement provides a roadmap for continued improvement and innovation.